Senior Design Project Proposal

SCOPE

Topic: Traffic Congestion & redirection prediction using Hybrid Quantum Machine learning

Guide: Prof. Karthika Natarajan

Team Members:

Jayanth Srinivas Bommisetty - 22BCE9512
Yaswanth Unguturi - 22BCE9538
Sushma Pidige - 22BCE9961

ABSTRACT

Rapid urbanization has led to a significant increase in traffic congestion, resulting in longer travel times, fuel wastage, and environmental pollution. Most existing traffic management systems focus on real-time monitoring and reactive control but lack effective mechanisms for predicting congestion trends and taking proactive decisions.

This project proposes a hybrid quantum–classical traffic congestion prediction system that combines deep learning–based perception with Quantum Machine Learning (QML) - based decision making. In the proposed system, YOLO is used to detect and count vehicles from traffic video feeds, enabling accurate extraction of traffic-related parameters such as vehicle density and lane occupancy. These extracted features are then processed and fed into a Quantum Machine Learning model to predict congestion levels and suggest traffic redirection or signal optimization strategies.

By separating perception and prediction into classical and quantum layers respectively, the system leverages the strengths of both approaches. The project aims to demonstrate the feasibility of applying QML to real-world smart transportation problems using quantum simulators, while maintaining a practical and scalable system design.

PROBLEM STATEMENT

Traffic congestion has become a major challenge in modern cities due to increasing vehicle population and limited road infrastructure. Conventional traffic control systems often rely on static signal timings or reactive adjustments based on current traffic conditions. Such approaches fail to anticipate congestion patterns and cannot efficiently adapt to sudden traffic surges.

Although classical machine learning techniques have been applied for traffic prediction, they often require large datasets and extensive feature engineering to model complex, non-linear traffic behavior. Additionally, most existing solutions focus only on traffic detection rather than predictive decision-making.

There is a need for an intelligent traffic management system that can not only detect traffic conditions accurately but also predict congestion levels and support proactive redirection decisions. Recent advances in Quantum Machine Learning provide a promising alternative for modeling complex correlations in structured, low-dimensional data. This project addresses the problem by integrating deep learning-based traffic perception with QML-based congestion prediction in a unified hybrid framework.

OBJECTIVES

The primary objectives of this project are:

- To detect and classify vehicles from traffic video streams using YOLO-based object detection
- To extract structured traffic features such as vehicle count, lane density, and congestion indicators
- To design a hybrid quantum–classical architecture for traffic congestion prediction
- To apply Quantum Machine Learning models for predicting congestion levels
- To suggest traffic redirection or signal adjustment decisions based on predicted congestion
- To compare the performance of QML models with classical machine learning baselines
- To demonstrate the practical feasibility of QML using quantum simulators

---

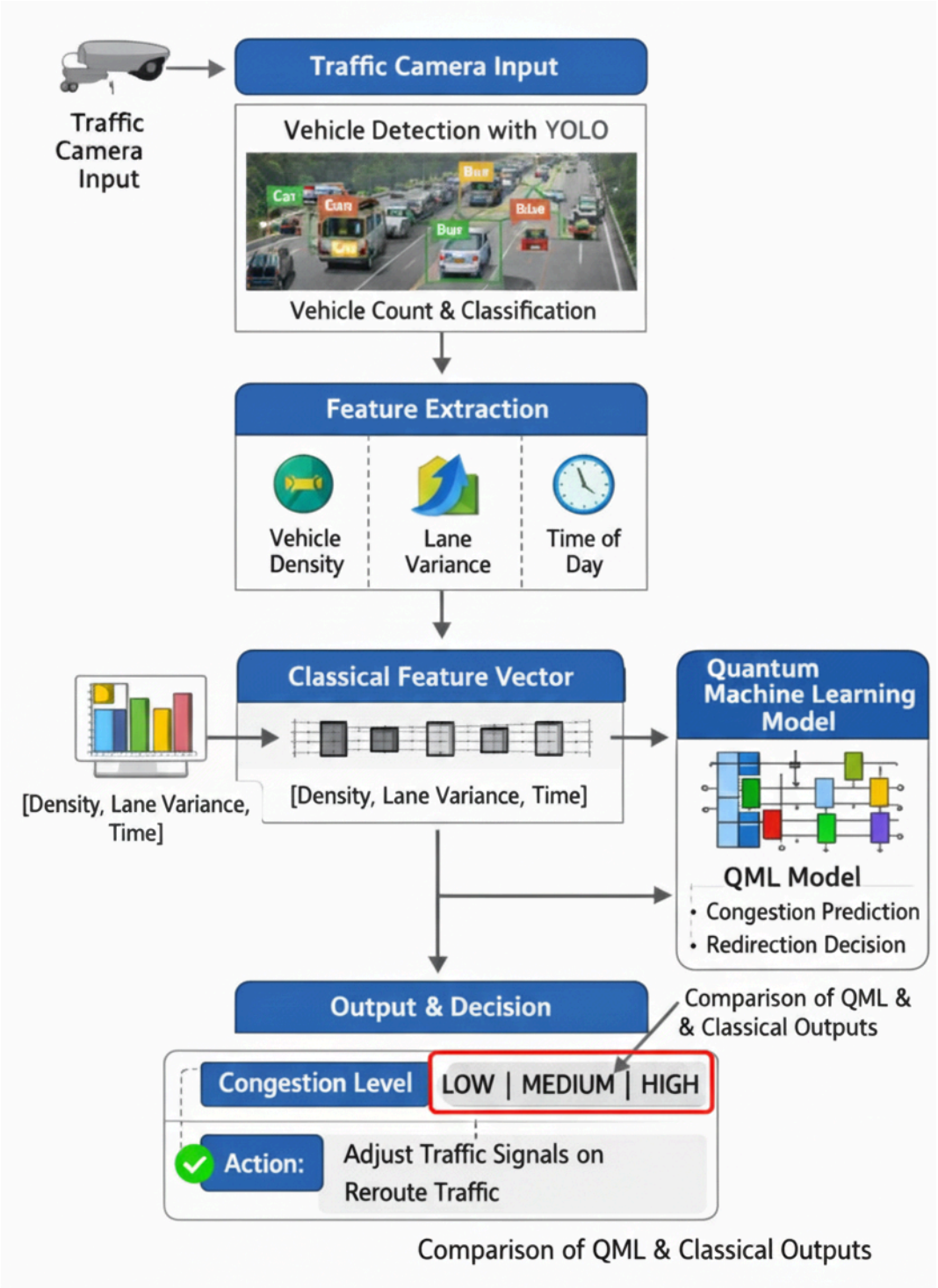
PROPOSED SYSTEM OVERVIEW

The proposed system follows a modular and layered architecture:

- Traffic Video Input
    - Traffic camera footage or recorded videos serve as input to the system.
- Vehicle Detection and Counting
    - YOLO is used to detect vehicles such as cars, buses, and bikes, and to count the number of vehicles per lane in each frame.
- Feature Engineering
    - The detected vehicle data is converted into structured numerical features, including traffic density, lane imbalance, and temporal indicators.

- Quantum Machine Learning Prediction
    - The extracted features are embedded into quantum states and processed using QML models to predict congestion levels and recommend redirection strategies.

- Output and Visualization
    - The system outputs predicted congestion categories (low, medium, high) along with suggested traffic control actions.



METHODOLOGY

The project adopts a hybrid quantum–classical methodology:

- YOLO performs high-dimensional visual perception tasks where classical deep learning is well-established.
- Feature dimensionality is reduced to ensure compatibility with QML models.
- Quantum Machine Learning is applied only to the decision-making layer, where structured, low-dimensional data is suitable for quantum embedding.
- QML models are implemented using quantum simulators due to current hardware limitations.
- Classical machine learning models such as SVM or Random Forest are used as baselines for comparison.

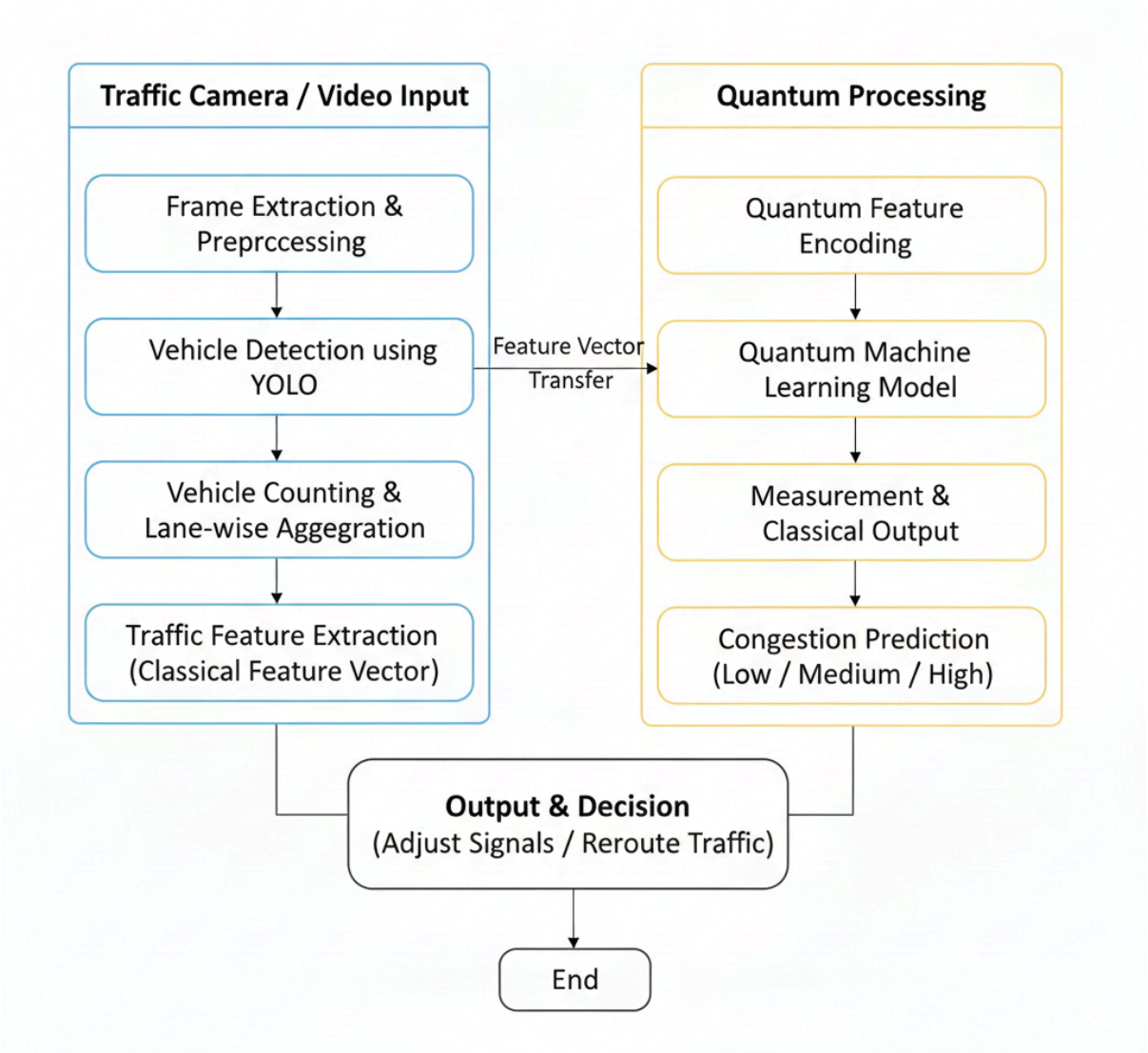
This approach ensures technical feasibility while exploring the potential advantages of quantum-based feature representation.

TECHNOLOGY STACK

- Python – Core programming language
- YOLO (v5/v8) – Vehicle detection and classification
- OpenCV – Video processing and frame extraction
- NumPy / Pandas – Data handling and feature processing
- Quantum Machine Learning Frameworks – QML model implementation using simulators
- Classical ML Libraries – Baseline model comparison

SYSTEM FLOW

1. Input traffic video is provided to the system
2. YOLO detects and counts vehicles in each frame
3. Traffic features are extracted and normalized
4. Features are embedded into a quantum model
5. QML predicts congestion levels and redirection decisions
6. Results are displayed to the user



EXPECTED OUTCOMES

- Accurate vehicle detection and traffic density estimation
- Effective prediction of traffic congestion levels
- Demonstration of QML applicability in traffic decision-making
- Performance comparison between quantum and classical models
- A scalable hybrid architecture for intelligent traffic systems

---

CONCLUSION

This project presents a hybrid quantum–classical approach to traffic congestion prediction by combining deep learning-based perception with Quantum Machine Learning-based decision making. By clearly separating the roles of YOLO and QML, the proposed system avoids unrealistic assumptions while exploring emerging quantum techniques in a practical context.

Although the system is implemented using quantum simulators, it serves as a strong proof-of-concept for future quantum-enhanced intelligent transportation systems. The modular design allows future extensions such as real-time deployment, integration with smart traffic signals, and experimentation with advanced quantum models as hardware matures.